

National Institute of Business Management School of Computing and Engineering Course work Assessment Announcement Sheet			
Course Name	HDSE		
Module Name	Data Management 2		
Batch			
Learning Outcomes Covered (Mention according to the Module Descriptor)	1. Identify the fundamentals of Oracle Database Server / SQL Server		
	2. Explain PL/SQL programming Techniques.		
	3. Apply Oracle for Database administration		
	4. Use Non-Relational Databases		
Assessment CW No	1		
Assessment Mode	Individual	Group (if it is group mode only)	
		Group Size	Grouping Criteria
Assessment Type	Software Presentation VIVA		
	If other specify		
Hand in Date Time			
Hand out Date Time			
Submission Details (Format and Location)	GitHub link of the project		
Plagiarism Acceptance Level	AI generated contents will obtain 0 marks.		
Assessment CW Description			
Note: This a take home assessment Description: You are a database developer working on a central management portal called "LifeLine Connect" for a regional blood bank network. The platform connects voluntary donors, hospital blood banks, and organizing committees to facilitate blood donation camps, track blood inventory, and manage donor eligibility. The system offers a wide range of functionalities, including donor registration, camp scheduling, blood inventory tracking, and donation history logging. • Design the ER or EER diagram. • Develop a simple Web or Enterprise Application for the LifeLine Connect system using any suitable programming language or framework. • Design and implement a relational database schema using Oracle Database.			

The database must support core system operations, including,

- Blood donation camp catalogue & venue management
- Donor registration & health eligibility tracking
- Blood inventory & blood group unit management
- Hospital requests & blood distribution processing
- Staff & volunteer assignment tracking
- Populate the database with sufficient sample data to simulate a realistic application scenario.
- Write and implement **PL/SQL programs** (procedures, functions, cursors, and/or triggers with exception handling) to perform database operations and business logic.

Your PL/SQL implementation should include at least five (05) business reports.

Examples:

- Retrieving total blood units collected by blood group across different donation camps
- Calculating blood inventory levels and identifying expiring units within a given time period
- Generating individual donor eligibility and donation history summary reports

MongoDB Integration

- Implement **MongoDB** as a complementary NoSQL database to handle unstructured campaign media, donor feedback, and emergency broadcast messages.

MongoDB collections should capture:

- Camp awareness materials, donor medical guideline documents, and campaign promotional media in flexible formats
- Donor experience reviews, ratings, and open-ended feedback for completed donation camps
- Emergency blood appeal notices, donor community discussion threads, and Q&A posts
- Ensure data is stored in a schema-flexible format to support dynamic content and community engagement.

Demonstrate queries in MongoDB to:

- Retrieve all feedback and reviews for a specific blood donation camp or venue
- Identify top-rated donation camps based on aggregated donor feedback

- Search discussion threads or emergency appeals for specific keywords or required blood types
- Integrate MongoDB with the application layer to provide real-time updates and interactive communication channels for donors and camp organizers.

Deliverables:

- Presentation
- Software Application with Database Implementation

Marking Rubric

Criteria	Weight	0% – 39%	40% – 49%	50% – 59%	70%+
Front-end Development + ER Diagram	20%	Very poor or missing front-end, ER diagram incomplete or incorrect	Basic front-end with minimal functionality, ER diagram partially correct	Functional front-end covering most operations, ER diagram mostly accurate	Fully functional, user-friendly front-end, ER diagram complete, normalized, and clearly represents all entities
Database Implementation (Oracle)	15%	Incomplete DB implementation, fails to meet requirements, no PL/SQL usage	Few operations (1–2) implemented with PL/SQL, limited security mechanisms	Two-thirds of operations implemented, some security mechanisms, acceptable backup plan	All operations implemented with PL/SQL, correct schema, strong security and backup mechanisms
MongoDB Incorporation	15%	No MongoDB implementation	Minimal MongoDB requirements captured, weak queries	MongoDB implemented with requirements, some queries demonstrated, partial integration	Correct MongoDB incorporation, requirements fully captured with standard queries
Reports & Business Logic (PL/SQL)	15%	No meaningful reports, poor exception handling	Few reports generated, limited exception handling	Reports implemented, acceptable exception handling	Comprehensive reports, robust PL/SQL logic, advanced exception handling
Completeness of the Project	5%	Incomplete, major components missing	Less than half completed	More than half completed but missing some features	Fully complete, all requirements achieved
Presentation & Viva	20%	Poor understanding; weak communication	Average understanding; average communication	Good understanding, clear communication	Excellent understanding, confident communication, demonstrates deep knowledge
Integration & Innovation	10%	No integration between Oracle and MongoDB	Minimal integration, weak linkage between systems	Partial integration, some cross-system use cases	Strong integration, innovative use of Oracle + MongoDB for hybrid analytics